COURSE : DATA ANALYTICS & DATA SCIENCE
ASSIGNMENT NO : 01
NAME : MILDRED LEDWABA
EMAIL : Makgolasi@gmail.com
CELL : 0832530574

Q1 What is a Database?

→ It is an electronic system used to collect information/data so that it can be easily accessed, analyse and draw information from

→ It enables companies to make decisions track records like customer sales, trends, demands and loses.

Schools → Learners data, reports, parents information birth certificates etc.

Liabrarties → Different Categories of books, authors, borrowers and return track records

Q. What is Data Warehouse?

→ It is a Central giant place where multiple data bases is Stored

Database - focus on Current, real-time data
Datawarehouse - focus on historical data collected from many sources.

→ It helps them make better, faster decisions. Track performance overtime.

③ What is a data lake?

-> It is a central storage system that holds massive amounts of raw data in its original format.

Datawarehouse -> Information is structured, cleaned and processed.

Data lake -> Information is raw, structured, unstructured and semi structured.

-> It allows them to store everything first and figure out later what insights they might want to use

-> It gives them flexibility for data scientists training machine learning models because they experiment with raw data.

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④ What is a Data Mart?

-> It is a smaller, focused piece of a data warehouse that contains tailored data of a specific department team or business line.

-> It is built directly from ~~your~~ central data warehouse The warehouse feeds cleaned, consistent data down to each mark and this keeps everything sync.

-> Data mart gives each team just the data they care about, so queries are faster and reports are simpler.

5. Structured, Semi-Structured and Unstructured.

Structured data -> Neat and tidy, spreadsheet application form.

Semi-Structured data -> Its flexible, has labels to group data, rows can vary. Emails.

Unstructured data -> No consistant format, no internal organization : Audio, CCTV Footage.

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6. Data Modeling

-> A process of creating a visual blue print for how data will be stored, organized and it defines what data you need, how tables connect and what rules apply.

-> It avoids chaos later, no duplication of data, It gives developers, analysts one shared diagram to agree on.

Conceptual Data model - High level business concepts and relationships - Online product.

Logical Data Model -> Details like attributes, keys data types, but still tech-agnostic.

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7. Data Mining?

-> The process of discovering patterns, trends and useful insights from large datasets using statistics

7.2 Companies use data mining to turn raw data into decisions that make or save money.

- Retail recommendation - Takealot, mining purchase behaviour history and browsing patterns.

Fraud detection -> Banks mining millions of transactions to learn normal spending patterns of each cardholder and flag fraud.

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8 What is Database Management System.

-> Its a tool used to interact with data base allows one to handle storage, queries.

MySQL - Open-source relational database man System, widely used to store and manage structured data.

PostgreSQL - "The world's most advanced open source ~~data~~ database".

Oracle Database -> high-performance, multi-model database management system, used for online transaction processing data warehousing

Mongo Database -> Open-source NoSQL document database that stores data in flexible JSON-like BSON format documents rather than tables.

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⑨ What is ETL (Extract, Transform, Load)

-> Pipeline process used to move data from source systems into a data warehouse or data lake so it can be analyzed.

-> Extract: pull raw data from source system into usable data.

-> Transform: Clean and reshape the data to make it useful - fixing errors, joining tables

-> Load: put the cleand, transformed data into the target system, usally a bw like snowflake.

-> It is important because it ensures reports are accurate and consistent and combines data from different systems into one place.

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⑩ Common data storage formats used in analytics

-> ~~GSV~~ CSV - Text, Raw-based-Common approach for small datasets and simple applications

-> JSON - Designed to store, manage, and query sem-structu documents, offering flexible schemas over rigid rational models.

-> Parquet -> designed for high-performance analytical process It is not a database itself but rather a file format that serves as a foundation for modern datalakes

(11) What is (AI) Artificial Intelligence.?

→ Is the field of computer science focused on building systems that can perform tasks which normally require human intelligence.

→ Google Maps → Brilliant at calculating the the fastest route using traffic data.

→ ChatGPT → , Spam filters, trained to follow an instruction in a promt and provide a detailed response.

(12) What is Machine Learning (ML)

→ ML → Is a subset of AI where systems learn patterns from data to make predictions or decisions.

ML vs Traditional Programming

Traditional → you write rules

ML → You give data and desired outputs → system learns rules.

Supervised → Uses labeled data : input + correct answer
Predict a known outcome.

Unsupervised → Uses unlabeled data : input only
· Find hidden patterns / groups

Reinforcement → Learns by ~~the~~ trial and error with rewards/penalties
· Maximize total reward over time

(13) What is Deep Learning

- A subset of machine learning, ~~based on artificial~~ learning, based on artificial nueral networks with multiple layers that enables computers to learn complex patterns from vast amounts of data without explicit programming.

- Neural Networks: Loosely inspired by the human brain.
  - Input layer - takes raw data: pixel of an image, words in a sentence.
  - Hidden layers → each learn features: Layer 1 might learn edges, layer 2 learns shapes, Layer 3 learns face.
  - Output layer - gives prediction
  - During training, the network adjusts internal weights to reduce errors. This is why it needs lots of data and computing power.

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(14) What is Natural Language Processing (NLP)

- Is the branch of AI that helps computers understand, interpret, and generate human language.

- Voice Assistants - Siri, Alexa - it understands intent and acts.
- Autocomplete & Grammar Check: Gmail fixing your writting
- Chatbots & Translation: Customer support chat or Google.

(15) What is a Large Language Model (LLM)

-> Is a type of deep learning model trained on massive amounts of text data to understand and generate human language.

Standard ML Model -> Trained for one narrow task: predict house prices, classify spam.

Large Language Model -> General-purpose for many language tasks without retraining.

- GPT-4 / GPT-5 (OpenAI): Powers Chat GPT.
  - Used for writing, tutoring, coding help
- Muse Spark (Meta): The model powering Meta AI
  - Used for conversational assistance, creative writing, image generation prompts.

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(16) What is Generative AI

-> Is a category of AI that creates new content rather than just analyzing existing data. It generates text, images, audio, video.

-> Trained on huge datasets -> The data model studies millions of examples: images, text, audio.

-> Learns patterns - It figures out structure, style and relationships.

-> Generates new output -> when prompted it uses those learned patterns to predict and create new content one piece at a time.

(17) What is a training Dataset

-> Is the collection of examples used to teach an AI model. It's the 'study material' the model learns from.

-> Its important because models learn patterns from data. The rule is "garbage in, garbage out"

-> If data is wrong, incomplete, or unrepresentative the model learns the wrong lessons.

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(18) What is Computer Vision.

-> Is the field of AI that enables machines to interpret and understand visual information from the world: images and videos.

-> Machines dont "see" like humans. To a computer an image is just a grid of numbers representing pixel color values.

-> Facial recognition for phone unlock: phone camera takes you face image.

-> Medical Imaging: Doctors use CV tools to analyze X-rays, MRIs and CT scans.